

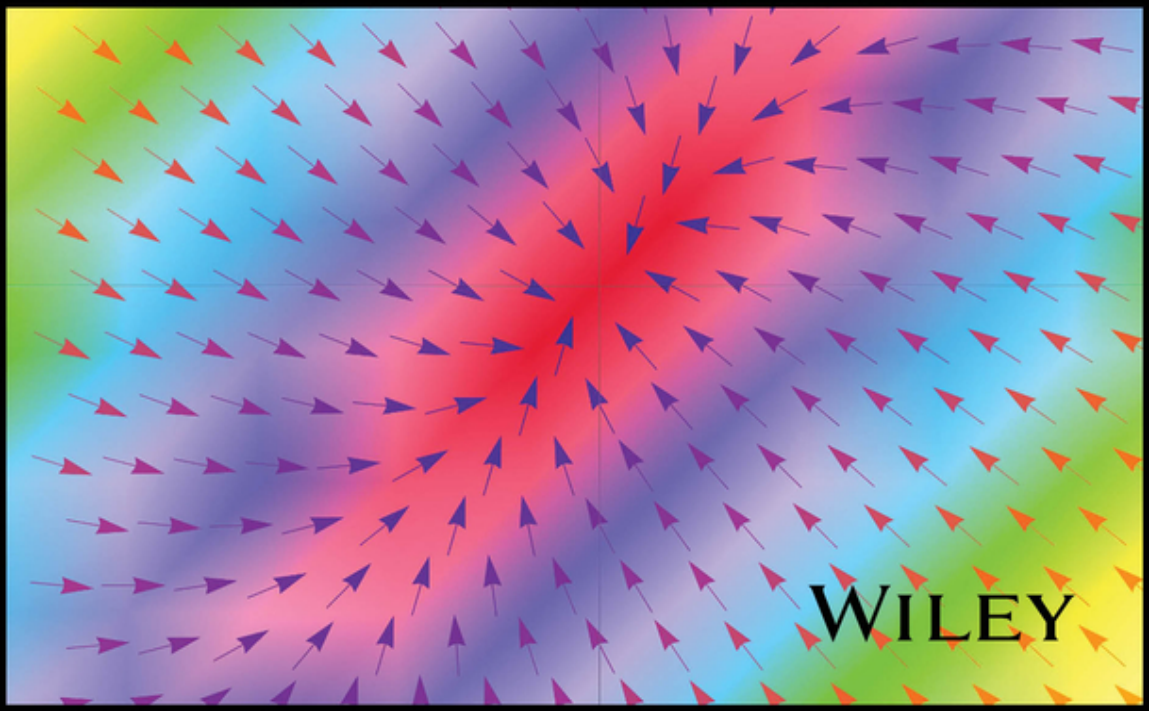
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SECOND EDITION

FIXED INCOME RELATIVE VALUE ANALYSIS

A Practitioner's Guide to
the Theory, Tools, and Trades

**DOUG HUGGINS
CHRISTIAN SCHALLER**



WILEY

Fixed Income Relative Value Analysis

Second Edition

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**A Practitioner's Guide to the
Theory, Tools, and Trades**

DOUG HUGGINS
CHRISTIAN SCHALLER

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This edition first published 2024

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Library of Congress Cataloging-in-Publication Data is Available:

ISBN 9781394189083 (Hardback)

ISBN 9781394189090 (ePDF)

ISBN 9781394189106 (ePub)

Cover Design: Wiley

Cover Image: Courtesy of Doug Huggins and Christian Schaller

Set in 10/12pt STIX Two Text by Straive, Chennai

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Preface to the Second Edition

Some important changes have occurred in the fixed income markets since we published the first edition of *Fixed Income Relative Value Analysis* in 2013, many of these due to the eventual policy response to the great financial crisis of 2008–2009: new reference rates, increasing default risk of governments, more regulation and capital constraints.

NEW REFERENCE RATES

Probably the most important effect of the great financial crisis from the viewpoint of relative value analysts and hence of this book was the transition away from LIBOR to other reference rates. After central banks largely disintermediated the interbank lending markets¹ that had been the lifeblood of the money markets, it was only natural that banks would prefer to borrow from and lend to a central bank rather than another commercial bank. But as a result of this change, the liquidity in the interbank market declined considerably, casting suspicion on interbank lending reference rates, such as LIBOR. The final straw for the LIBOR market was the rate rigging scandal that encompassed a number of large banks, starting in 2008. In response to these issues, central banks in a number of jurisdictions pushed market participants away from LIBOR and toward reference rates of their own design, such as SOFR in the US. In fact, the move to SOFR was the impetus for our recent book, *SOFR Futures and Options* (Huggins and Schaller, 2022).

Since SOFR differs from LIBOR in a number of important ways, the SOFR swaps market differs from the LIBOR swaps market in a number of important respects. For example, LIBOR references unsecured transactions, whereas SOFR references secured transactions. This change alone has a material effect on valuations, particularly on the relative valuations between swaps and bonds.

¹Central banks not only have disintermediated the money markets in some jurisdictions, they have also disintermediated some of the bond markets. For example, the buying of JGBs by the official sector in that market has reduced the liquidity of the world's largest bond market at times to a trickle.

On one hand, the multitude of reference rates (Chapter 11 provides an overview) for swaps has complicated the already complex relationships between asset, basis, and credit default swaps further. On the other hand, the transition from LIBOR to SOFR has eliminated the unsecured–secured basis in the asset swap spreads of government bonds, which has had several welcome effects:

- Asset swap spreads versus SOFR can be modified to become suitable rich/cheap indicators for all bonds worldwide (Chapter 17). We have therefore reviewed the strong statement of the first edition that asset swap spreads should never be used to assess the RV between bonds. While it is still true for swap spreads versus LIBOR and OIS, par asset swap spreads versus SOFR can be modified to become a reasonable alternative to fitted curves.
- Asset swap spreads versus SOFR can rather easily be linked with the credit risk of the sovereign issuer. Hence, the first hurdle facing an arbitrage equality between swap spreads and CDS, which occupied many pages of the first edition due to the correlation between credit risk and the unsecured–secured basis, now has a straightforward solution (Chapter 16).

A less welcome consequence of the transition to SOFR are the structural breaks in time series, for example, for cross-currency basis swaps between EURIBOR and USD LIBOR, which switched to USD SOFR. This is a general problem for analysts requiring long-term time series as input variables for their models. For the most basic of those time series, interest rates, we propose the solution of using constant maturity par yields from a fitted curve on government bonds (Chapter 8) rather than swap rates. And it is also a specific problem for this book, written shortly after the transition and hence with too little data for swaps with new reference rates to present meaningful case studies. We have therefore decided to keep many of the case studies of the first edition, which cover the subprime and euro crises and thus remain of interest, though this means that we need to carry some “historical ballast” and speak from time to time of (basis) swaps with USD LIBOR as the reference rate. Apart from these case studies, we have aimed for a complete update to the current reference rate situation.

INCREASING DEFAULT RISK OF GOVERNMENTS

The implementation of zero or even negative policy rates posed a challenge to analysts – as does its unwinding, which occurs at the same time as budget deficits and bond issuance soared, in part as a consequence of the fiscal

response to the COVID-19 pandemic. An immediate implication for modeling government bonds is that the formerly common simplifying assumption that certain government bonds were default-free is less justifiable these days. As a result, incorporating credit considerations into the analysis of government bonds has become increasingly important. We have responded to this development by adding a term for credit difference to the term for funding difference in our swap spread model. While the sovereign CDS was already treated extensively in the first edition, it remained an external addition to the relationship between bonds and swaps, into which it has now been integrated (Chapter 12).

On the other hand, moves in rates and volatility even to unprecedented levels are the sorts of changes for which most good relative value models are well-suited or can be adapted. We provide a study of the impact of zero-interest rate policy on PCA eigenvectors in Chapter 3 and offer some remarks about Shadow Rate Models in Chapter 6. If those were the only changes occurring in the fixed income markets, the need for a second edition of *Fixed Income Relative Value Analysis* would be less clear.

But the move from LIBOR to overnight reference rates, non-negligible default risk of government bonds, and tighter regulation are the sorts of changes that require us to review and partly modify our approaches. The basic economics underlying our models are the same. They are still predicated on the principles that arbitrage opportunities are unsustainable and that instruments with similar risks should be expected to generate similar returns (Chapter 1). But the application of these principles via models needs to change as the structure of the market changes. And that's the primary motivation behind this second edition of the book.

REGULATION AND CAPITAL CONSTRAINTS

The Basel III process has culminated in quite a number of additional regulations designed to improve the safety and soundness of the world's banks, especially the largest institutions that tend to dominate the fixed income markets. One effect of these regulations is that bank balance sheets face many more constraints now than they did a decade ago. And the effect of these constraints is that the relevant cost for a bank contemplating a new position is no longer the marginal cost of funds in the market, such as interest rates paid on retail or commercial deposits. Rather, it's the *shadow cost* of the least onerous constraint currently binding the bank's balance sheet. For example, if a bank needs to shed an asset currently generating a return on capital of 7% in order to make room for another asset, then 7% is the relevant hurdle for the new asset, even if retail deposits cost the bank much less.